국가데이터처 통계인재개발원 | 보도자료 | APEC 2025 KOREA

보도시점 2025. 10. 29.(수) 12:00 배포 2025. 10. 29.(수) 08:30

2024-2025 국제 통계포스터 대회 수상 결과 발표

- 대한민국, 초등부 및 중등부 1위, 대학부 장려상 수상 -

국가데이터처(처장 안형준)는 2024-2025 국제 통계포스터 대회(ISLP* Poster Competition)에서 대한민국 대표팀이 **초등부와 중등부 1위를 차지하고, 대학부에서는 장려상**(Honourable Mention)**을 수상**하는 성과를 거두었다고 밝혔다.

* ISLP(International Statistical Literacy Project): 국제통계교육프로젝트

국제 통계포스터 대회는 초·중·고 및 대학생 등 **학생들의 통계와 데이터를 활용한 창의적 문제해결 능력을 평가**하기 위해 국제통계기구(ISI) 산하단체인 통계교육국제협회(IASE)가 주관하는 **국제행사로 금년은 총 37개국이 참가**하였다.

국가데이터처는 학생들의 데이터 활용 역량 강화를 위해 매년「전국학생통계활용대회」를 개최하고 있으며, 금번 2024-2025 국제 통계포스터 대회에서 초등부, 중등부에서 1위를 차지한 팀은 **「2024년 전국학생통계활용대회」에서 각각 초등부 대상과 중등부 금상을 수상한 팀**으로, 국가데이터처가 국제대회 참가를 지원하였다.

초등부에서는 진천상신초등학교 임도윤, 박태용, 이도영 학생팀이「선택해야 한다면 무엇을 하시겠습니까? (If You Had to Choose, What Would You Do?)**」라는 주제로** '학생들의 선택 행동을 통계적 관점에서 분석한 작품'으로 **1위를 수상**하였다.

중등부에서는 한국외국인학교(판교캠퍼스) **노정민 학생이「AI 음악에 대한 인식과 편견** (Perception and Bias Towards AI Music)**」이라는 주제로** 'AI가 만든 예술 작품에 대한 사회적 인식과 편견을 분석'하여 **1위를 수상**하였으며,

대학부에서는 경북대학교 김현진, 박승정, 이동근, 오유진 학생팀이「싱크홀 발생 원인과 위험도를 예측한 회귀모형의 지리적 분석 (Geographical Analysis of Regression Model with Application to Sinkhole Occurrence)**」연구 결과가 장려상을 수상**하였다.

안형준 국가데이터처장은 **“금번 우리나라 학생들의 수상을 축하하며, 참가 학생 모두의 노고에 감사를 전한다. 또한, 학생들이 통계와 데이터를 보다 창의적이고 적극적으로 활용할 수 있도록 지원과 노력을 아끼지 않겠다”**고 약속하였다.

[붙임1] 「국제 통계포스터 대회 및 전국학생통계활용대회」 개요
[붙임2] 「2024-2025 국제 통계포스터 대회」 대한민국 수상작

담당 부서	통계인재개발원 교육기획과	책임자	과 장 차진숙 (042-366-6101)
		담당자	사무관 박민식 (042-366-6151)



붙임 1 국제 통계포스터 대회 및 전국학생통계활용대회 개요

□ 국제 통계포스터 대회 개요

국제 통계포스터 대회(ILSP)는 국제통계기구(ISI) 산하단체인 통계교육국제협회(IASE) 주관으로 진행하는 **'국제 통계능력 향상 프로젝트의 일환'으로 실시**

○ **(개최주기)** 2년

○ **(참가자격)** 전 세계 초·중·고등학생 및 학부생

○ **(팀 구성)** 1~5명으로 구성

○ **(작품요건)** 자유주제, 통계적 정보를 활용·분석하여 전달하는 통계포스터

○ **(심사방법)** 작품심사

○ **(시상내역)** 부문별(초・중・고・대학부) 1등, 2등, 3등 각 1팀, 장려상 3팀

○ **(수상혜택)** 국제대회 수상작은 국제통계기구(ISI) 세계 통계 총회에서 발표되며, 소정의 상금과 상장 수여

□ 전국학생통계활용대회 개요

○ **(주최 / 후원)** 국가데이터처 / 교육부 및 시・도 교육청

○ **(개최주기)** 매년

○ **(참가자격)** 초·중·고등학교 재학생 및 동일연령 청소년

○ **(팀 구성)** 3명이하 학생 및 지도교사 1명으로 구성

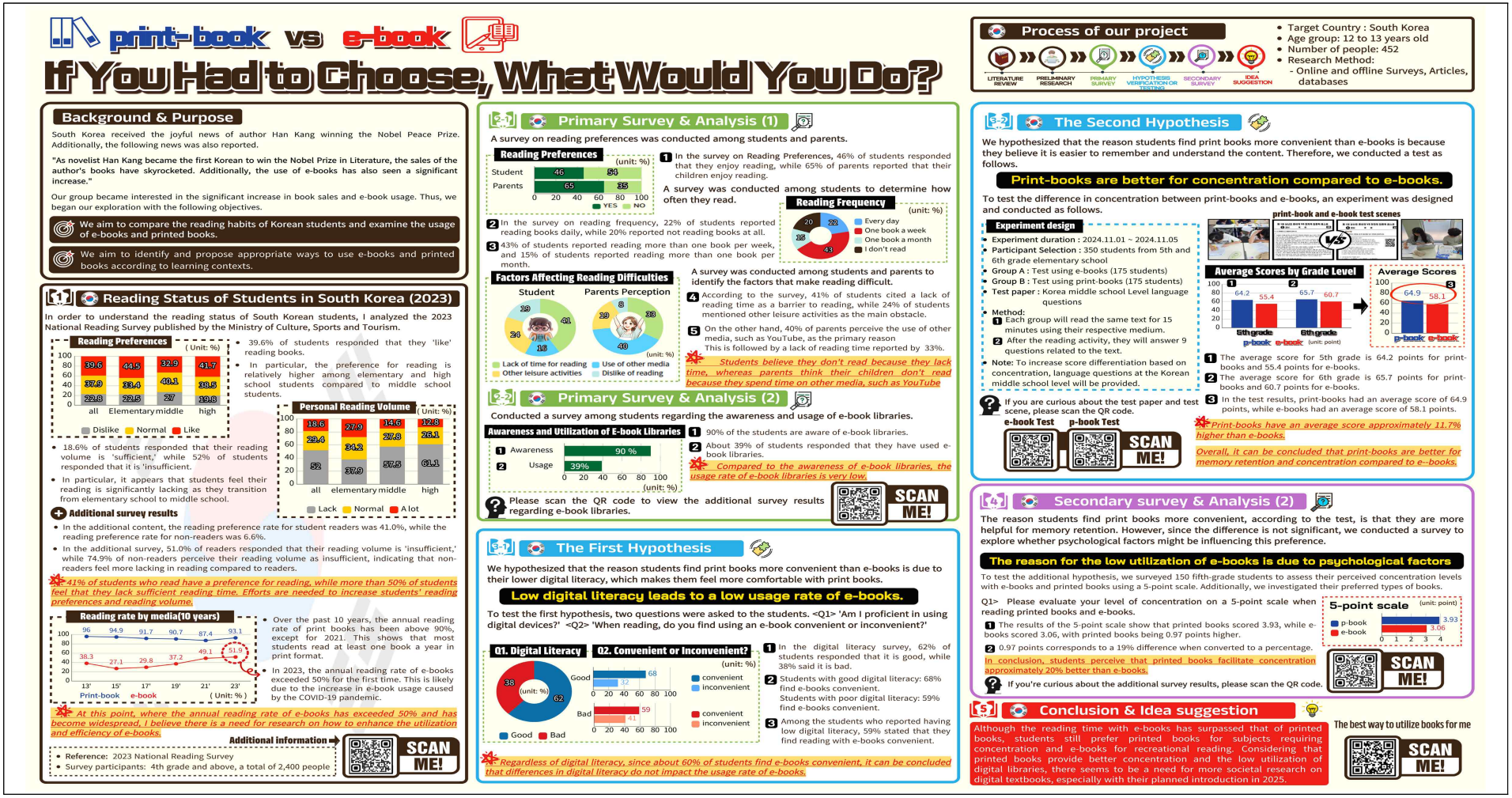
○ **(심사방법)** 작품심사 + 발표심사

○ **(시상내역)** 117개팀(대상, 금상, 은상, 동상, 장려상) 및 단체상 3개교

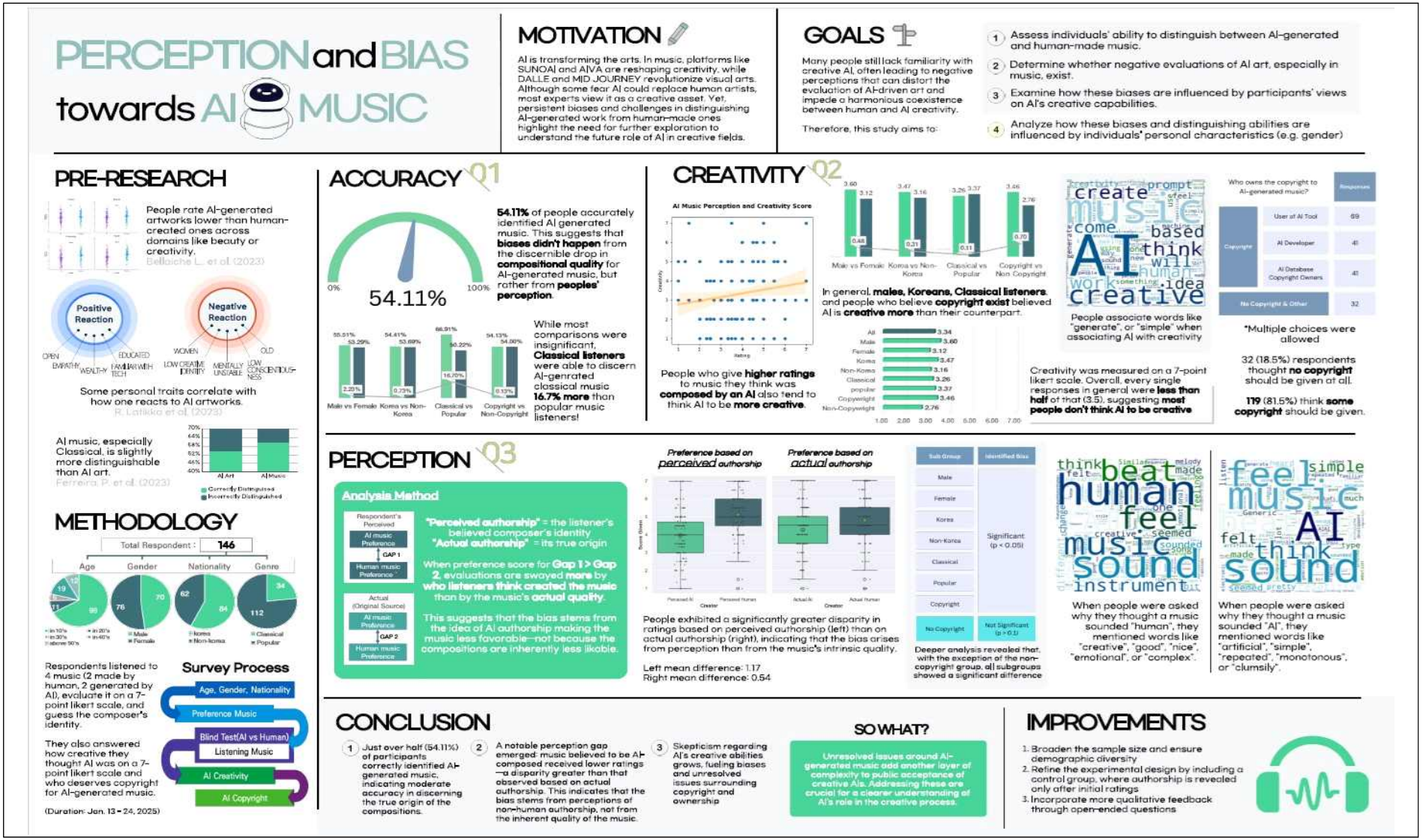
○ **(수상혜택)** 상격별 장학금 및 상장 수여

붙임 2 2024-2025 국제 통계포스터 대회 대한민국 수상작

□ 초등부 (Elementary School Division) 1위 : 진천 상신초등학교(임도윤, 박태용, 이도영)



□ 중등부(Lower Secondary Division) 1위 : 한국외국인학교(노정민)



□ 대학부(Undergraduate Division) 장려상 : 경북대학교(김현진, 박승정, 이동근, 오유진)

Geographical Analysis of regression model with application to Sinkhole occurrence

1. Introduction

A sinkhole is a hole formed when the upper layer of soil collapses due to a void created beneath the ground. In South Korea, frequent sinkholes have caused significant casualties and property damage, yet research remains limited. Internationally, sinkholes often result from the dissolution of limestone by groundwater, and extensive studies have been conducted on this phenomenon. However, since limestone areas in South Korea are mostly limited to the Gangwon Province, directly applying such findings is challenging. Additionally, prior studies have proposed installing Internet of Things(IoT) devices in areas susceptible to sinkhole formation, considering factors such as water leakage from supply and drainage systems or groundwater outflow. However, installing IoT devices in all regions is impractical.

2. Objectives

The goal is to predict the probability of sinkhole occurrences in order to prepare for and mitigate human casualties and property loss caused by sinkholes. However, the primary causes of sinkhole formation are likely to vary by region, making it challenging to analyze the entire country. Therefore, this study aims to predict the probability of sinkhole occurrences for each district in Seoul, the most densely populated city in South Korea. To develop a sinkhole occurrence probability prediction model that incorporates the artificial and environmental characteristics of Seoul, which have not been addressed in previous studies. Through this, the installation of Internet of Things(IoT) devices can be prioritized in high-risk districts.

3. Data

- 467 dongs in Seoul divided into districts(Eup, Myeon, Dong) based on the legal boundary
- Sinkhole accident data was obtained from the 'Underground Safety Information System', and a total of 83 accidents existed
- Sinkhole accident data from January 2020 to September 2024
- Flooding traces, traffic volume, building density, population density, presence of subway stations, and groundwater outflow were adopted as variables expected to affect sinkhole accidents. And there are three types of groundwater outflow: subway groundwater outflow, building groundwater outflow, and communication port groundwater outflow.

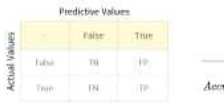
4. Analysis

A 467×467 distance matrix was constructed based on the administrative boundaries of Seoul's districts (Eup, Myeon, Dong). The matrix was defined such that if the centroid of a district lies within a 2km radius or directly neighbors another district, the corresponding value is 1; otherwise, it is 0.

During the analysis, it was identified that the traffic volume variable contained two missing values. Given that traffic volume is likely associated with neighboring districts, Moran' s I statistic was calculated using the W matrix to assess spatial autocorrelation. Therefore, the missing values were imputed using the average traffic volume of the adjacent districts.

The distance matrix was then multiplied by the independent variables to create spatial interaction variables, which were incorporated as predictors in the regression models. Various models, including Poisson regression, logistic regression, and the Zero-Inflated Poisson(ZIP) model, were applied for analysis.

Model performance was evaluated using Recall, and Accuracy, with higher values indicating better performance. Since accurately identifying areas where sinkholes occur is of primary importance, Recall was considered the most critical evaluation metric.

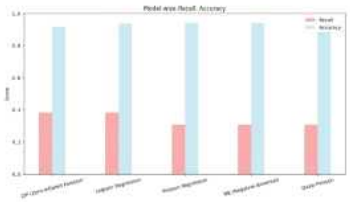


$$Recall = \frac{TP}{TP + FN}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

pmf of ZIP

$$P(Y_i = y_i) = \begin{cases} \pi_i + (1 - \pi_i)e^{-\lambda_i} & \text{if } y_i = 0, \\ (1 - \pi_i)\frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!} & \text{if } y_i > 0. \end{cases}$$

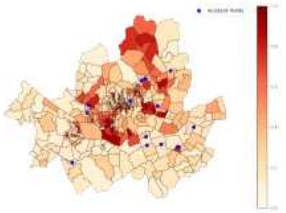


5. Result

This figure represents the result of smoothing using a weighted average, with a 7:3 ratio between Y and WY, where WY is the average of Y values from neighboring districts. The smoothing is applied to the results of the ZIP model. The 7:3 ratio was chosen because it yielded the best recall value during evaluation. The figure on the right visualizes the smoothing result, with blue dots representing sinkhole occurrences in 2024.

fitted model

$$\begin{cases} \log\left(\frac{P(Y_i=0)}{1-P(Y_i=0)}\right) = 0.0175 - 0.5034 \cdot \text{subway_bi} - 0.1652 \cdot \log(\text{building_den}) - 0.3915 \cdot \text{out_sum_sum_weighted} \\ \log(\lambda_i) = -1.4858 + 0.5790 \cdot \text{subway_bi} + 0.1043 \cdot \log(\text{building_den}) + 0.0611 \cdot \text{out_sum_sum_weighted} \end{cases}$$



Predictive Values

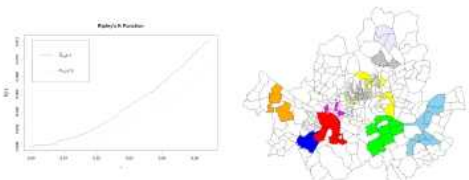
Actual Values	False	True
False	327	127
True	3	10

Recall : 0.769
Accuracy : 0.721

7. Suggestions

By the primary causes of sinkholes, water leakage from underground water pipes was identified, but data could not be used due to restrictions from local governments. If data on the age of water supply and drainage pipes could be incorporated into the study, the effectiveness of the research could be further enhanced.

Since the sizes of Seoul's districts vary significantly, it may be necessary to adjust for differences in spatial proximity. Based on Ripley's K analysis, spatial clustering was confirmed, suggesting the need to expand the analysis scope in future research using a clustering-based approach.



6. Conclusion

- The high-risk areas for sinkhole occurrence in Seoul identified by the ZIP model were Hwagok-dong, Jayang-dong, Yeoksam-dong, Geoyeo-dong, and Bangbae-dong.
- The risk of sinkhole occurrence increases with the presence of subway stations, higher building density, and higher frequency of groundwater leakage.
- Based on these results, it can be proposed to local governments to install Internet of Things (IoT) sensors in the above five areas for sinkhole detection.

8. Future work

Since sinkholes typically form as inverted cones, bowl shape or cylinder(Kim, 2017) and we have width, length and depth of each accident, the incident size of inverted cones can be expressed as:

$$\text{Sinkhole Volume} = \frac{1}{3} \times \left(\frac{\text{Width}}{2}\right) \times \left(\frac{\text{Length}}{2}\right) \times \text{Depth} \times \pi$$

A model predicting the scale of sinkhole incidents could be developed to alert local governments where the largest predicted sinkhole sizes are expected, urging them to take preventive measures.

Sources

SUNGHYOHYUN, & AHNSEJIN (2017). GIS Based Sinkhole Susceptibility Analysis in Karst Terrain: A Case Study of Samcheok-si.

Jae Heum Lee, Heyng Sub Lee, & In-Hwan Lee (2015). IoT-based Convergence Technology for Urban underground sinkhole Prediction.

Kim, C. (2017). A study on the causes and countermeasures for sinkholes in urban areas. Master's thesis, Graduate School of Construction and Industry, Kyonggi University, Gyeonggi-do, Republic of Korea.